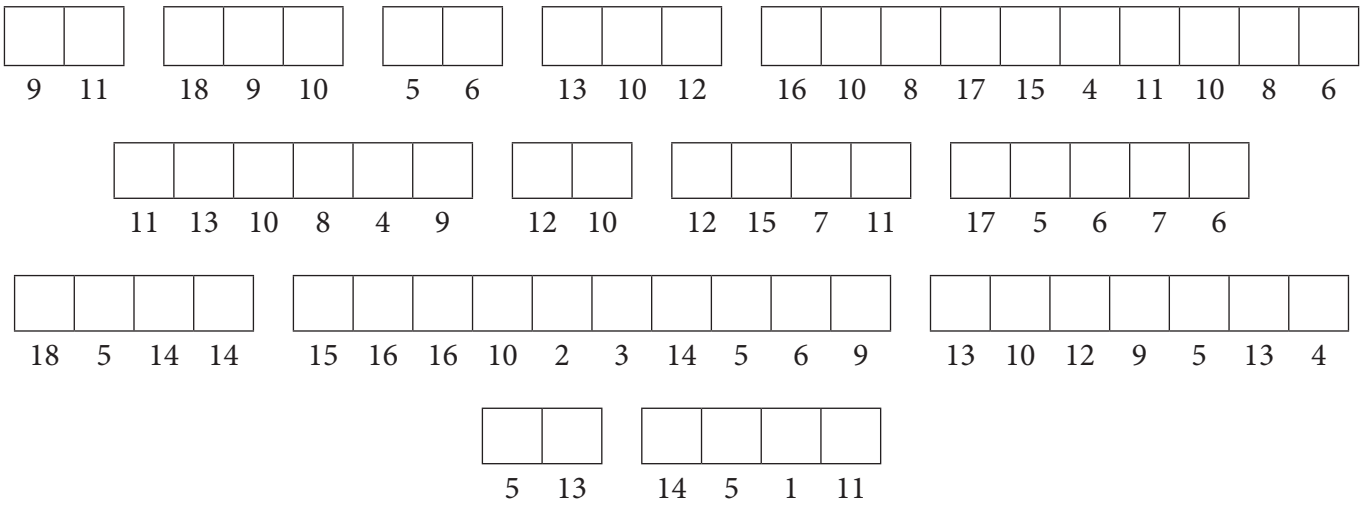
WORKSHEET

# Chapter 3: Absolute values




*Muhammad Ali*

## Questions

State the value of the following:

**1** $|5|$

**2** $|-2|$

**3** $|6-13|$

**4** $|4\times-7+2\times9|$

**5** $|-8|\times3|5|-|-48|$

**6** $-|42|\div|-3|+|-2\times5|$

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Solve the following equations to find $x$.

**7** $|4x| = 28$

**8** $|9x - 2| = 25$

**9** $|7 + 5x| = x + 29$

**10** $|6x - 11| = |25 - 3x|$

**11** $35 < |-5x|$

**12** $|2 - 8x| > 18$

**13** Solve $|6x - 3| \geq 45$

**14** Sketch the solution to Question 13 on a number line.

**15** Solve $|4x + 2| < -9 - 7x$

**16** Sketch the solution to Question 15 on a number line.

**17** Solve $|2x + 1| < |x + 4|$

**18** Sketch the solution to Question 17 on a number line.

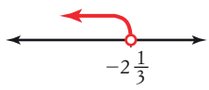
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## Solutions

**A** $x < -2\frac{1}{3}$

**C**




**E** $x < -7$ and $x > 7$

**F** 5

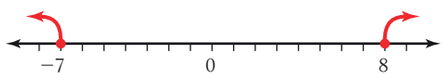
**G** 10

**H** $x = -6$ and $5\frac{1}{2}$

**I** 72

**K** $x = \pm 7$

**L**




**M** 2

**N** $x \leq -7$ and $x \geq 8$

**O** $x = -4\frac{2}{3}$ and 4

**P** 7

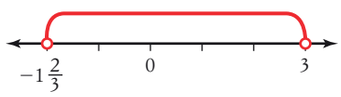
**R** $-1\frac{2}{3} < x < 3$

**S** $-4$

**T** $x < -2$ and $x > 2\frac{1}{2}$

**U** $x = -2\frac{5}{9}$ and 3

**W**




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